Proofs for Cylindrical Algebraic Decomposition

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Progress Report 2

Nonlinear Real Arithmetic

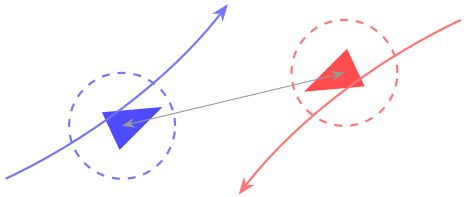
- **Nonlinear Real Arithmetic** (NRA): polynomial (in)equalities over $\mathbb{R}$

$$ax^2 + bx + c = 0 \wedge b^2 - 4ac < 0$$

- Question: is there a valuation for the variables that makes the formula **true**?

Applications

- Hybrid systems, e.g. collision avoidance



$$(x_1 - x_2)^2 + (y_1 - y_2)^2 \geq p^2$$

- Verification of programs with exact arithmetic

Nonlinear Real Arithmetic

- **Cylindrical Algebraic Decomposition** (CAD), Collins 1975: complete algorithm for NRA; even if the formula has quantifiers.
  - Doubly exponential in the number of variables
  - Algorithms with better bounds exist, but CAD wins in practice

Nonlinear Real Arithmetic

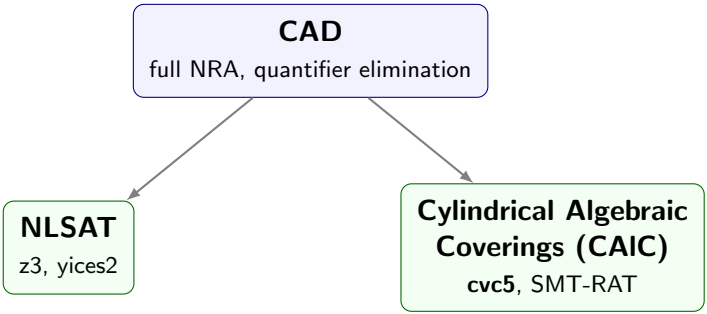
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  - Doubly exponential in the number of variables
  - Algorithms with better bounds exist, but CAD wins in practice

- Standard method in computer algebra systems





CAD-based Algorithms in SMT



faster for quantifier-free NRA

Trusting CAD

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+ No per-run overhead
– Every optimization must be verified

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Proof-producing implementation

+ Heuristic search stay unverified
+ Reuse an optimized solver as-is



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- Previously, we achieved both these goals for problems with a single variable, with a checker based on standard CAD.
  - The certificate consisted of just the roots of the polynomials in the problem
- Since then we developed a fine-grained proof calculus for this fragment based on CAIC, allowing cvc5 to produce more detailed proofs. It is now $3\times$ faster, solving 12 new problems.

Related Work

Approach	**Work**	**Kind**
Sums of squares	Harrison 2007	certificates (HOL Light)
Gröbner bases + sums of squares	Platzer et al. 2009	proof producing
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Univariate CAD	**Li et al. 2019**	**certificates (Isabelle/HOL)**
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No integration in SMT solvers, and no multivariate CAD so far.

Outline

- Introduction ✓
- Univariate CAD and Univariate CAIC
- Proof Reconstruction for Univariate CAIC

Univariate CAD and CAIC

Univariate CAD

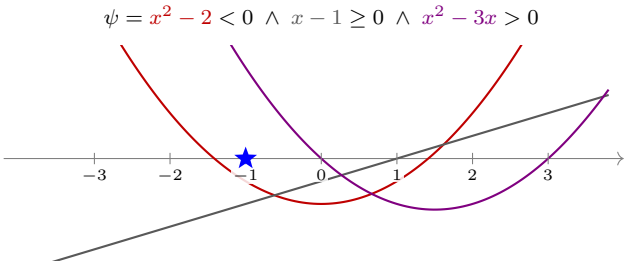
- How do we know if there is an $x \in \mathbb{R}$ satisfying the formula below?
- There is an uncountable number of candidates, so enumeration is impossible.

Example

$\psi = x^2 - 2 < 0 \ \wedge \ x - 1 \geq 0 \ \wedge \ x^2 - 3x > 0$
Candidates for x: $\mathbb{R}$

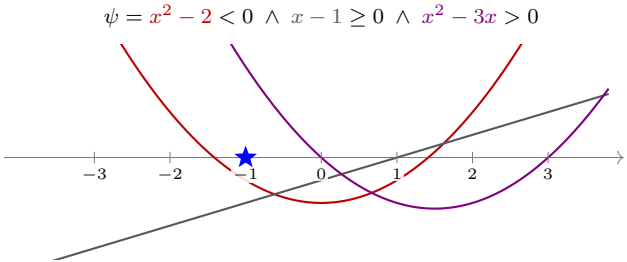
Univariate CAD

- When one value fails (e.g. -1), we can exclude more values based on two observations:



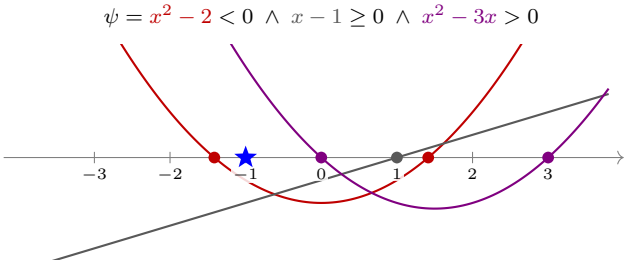
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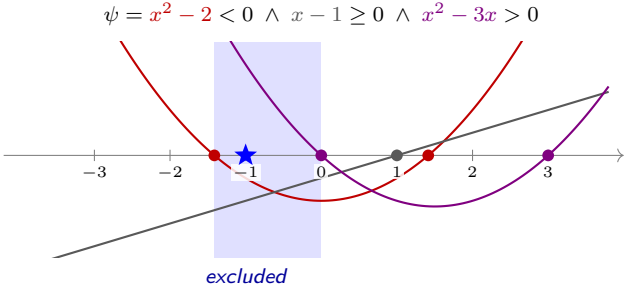
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Univariate CAD

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  - All assertions are about the sign of some polynomial
  - Polynomials only change their sign on one of its roots

- Thus, we can safely exclude the interval around -1 that doesn't contain any root.



Univariate CAD

- Our new set of candidates will be $\mathbb{R} \setminus (-\sqrt{2}, 0)$

- We choose a new point in this set and repeat, until either we find a satisfying point or we cover the whole line.

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Univariate CAD

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1: **function** UNIVARIATECAD(ψ)
2: Let $p_1, p_2, \cdots, p_n$ be all polynomials in ψ.
3: Let $r_1, r_2, \cdots, r_m$ be all the real roots of $p_1, \cdots, p_n$, sorted.
4: $\Gamma \leftarrow \mathbb{R}$
5: **while** $\Gamma \neq \varnothing$ **do**
6: Let $a \in \Gamma$
7: **if** $\psi[x \mapsto a]$ evaluates to true **then return** SAT
8: Let i be such that $a \in (r_i, r_{i+1})$
9: $\Gamma \leftarrow \Gamma \setminus (r_i, r_{i+1})$
10: **return** UNSAT

Univariate Cylindrical Algebraic Coverings

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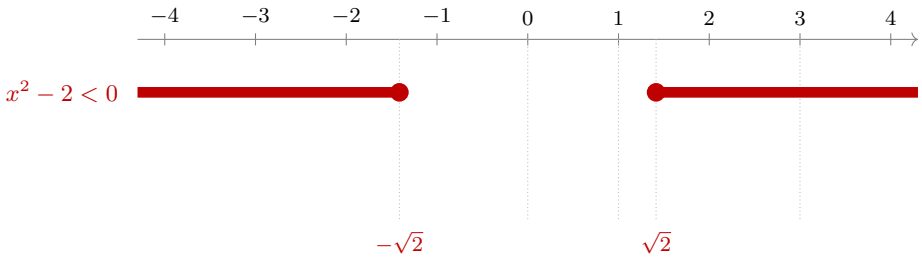
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- Coverings handles one constraint at a time: for each $p \bowtie 0$, compute the intervals where it is false, using only the roots of p

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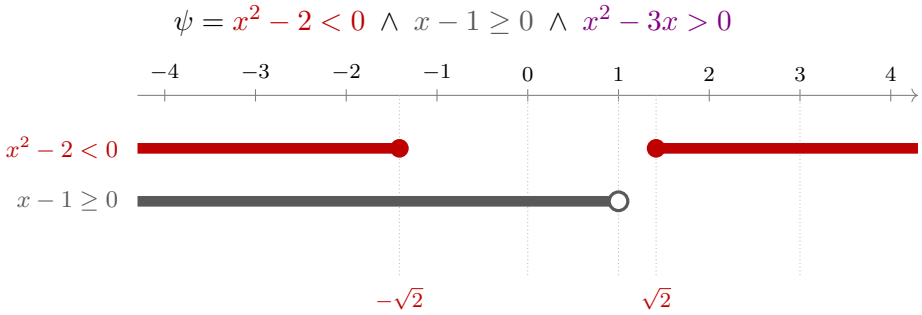
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- The coverings variant is designed for conjunctions of constraints, which is what the SAT solver gives to the theory solver
- Coverings handles one constraint at a time: for each $p \bowtie 0$, compute the intervals where it is false, using only the roots of p
- If these intervals cover $\mathbb{R}$, the formula is UNSAT; otherwise, any point outside them satisfies it

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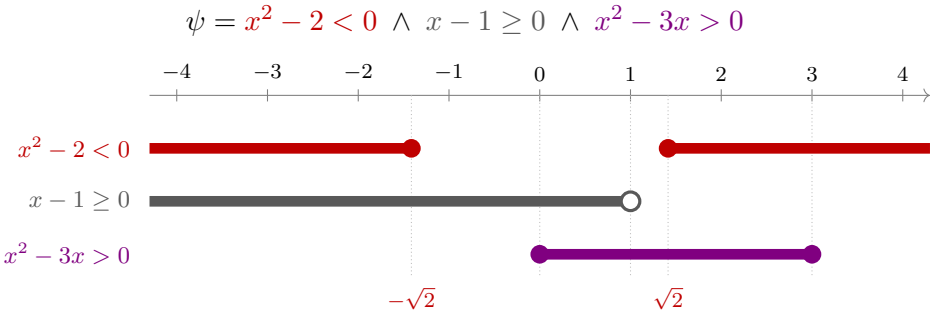
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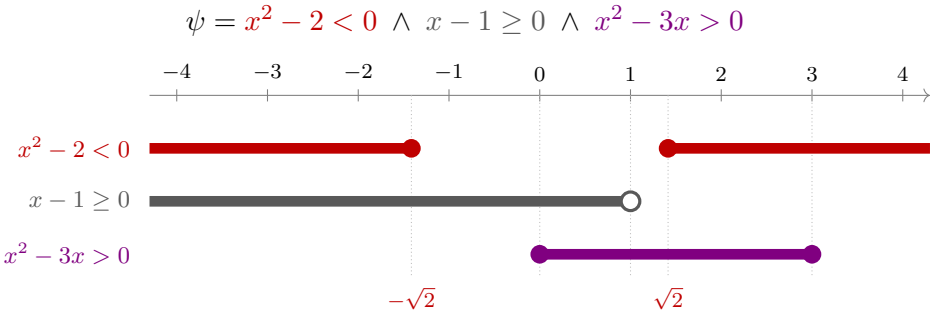
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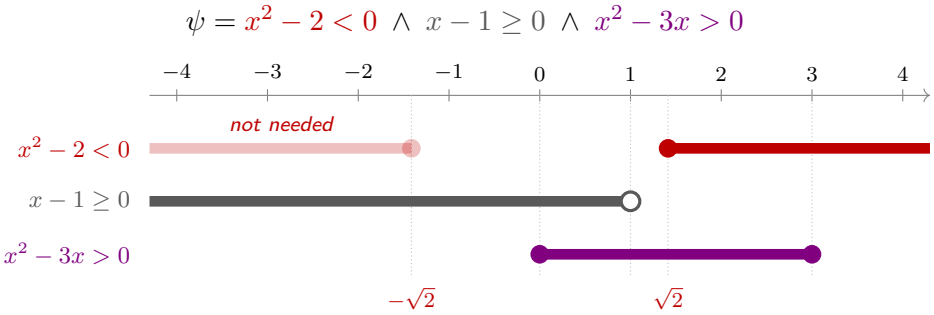


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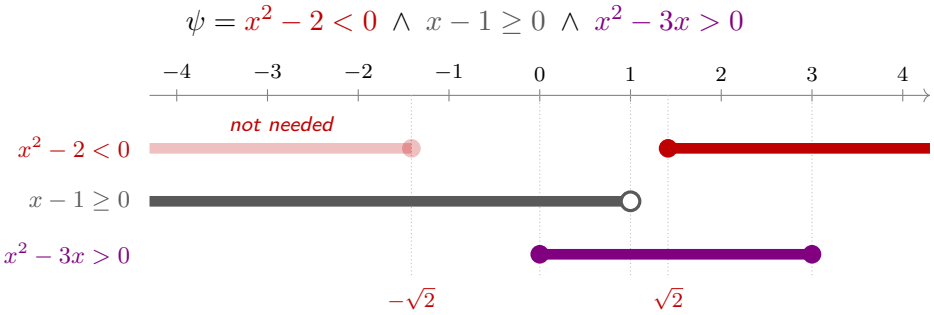
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- The intervals cover $\mathbb{R}$, so ψ is UNSAT
- Covering: $(-\infty, 1)$, $[0, 3]$, $[\sqrt{2}, +\infty)$; each interval involves a single polynomial
- No polynomial is evaluated at an irrational point. In contrast, CAD needs to evaluate $x - 1$ and $x^2 - 3x$ at $\pm\sqrt{2}$

Proof Production and Reconstruction

The Certificate

- We use the list of intervals (paired with the offending constraint) produced by cvc5 to generate a proof certificate.

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Example

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Original intervals:

$(-\infty, 1)$	$[0, 3]$	$[\sqrt{2}, +\infty)$
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	$x - 1 \geq 0$	$x^2 - 3x > 0$	$x^2 - 2 < 0$
After splitting:	$(-\infty, 1)$	$\{0\},\ (0, 3),\ \{3\}$	$\{\sqrt{2}\},\ (\sqrt{2}, +\infty)$

The Shape of the Proof

- We introduce proof rules for certifying that the variable must be in one of the intervals (as they cover the real line) and for excluding both point intervals and open intervals.

$x - 1 \geq 0$ $x^2 - 3x > 0$ $x^2 - 2 < 0$

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$x - 1 \geq 0$

$x^2 - 3x > 0$

$x^2 - 2 < 0$

COVER

$$x < 1 \ \vee \ x = 0 \ \vee \ 0 < x < 3 \ \vee \ x = 3 \ \vee \ x = \sqrt{2} \ \vee \ x > \sqrt{2}$$

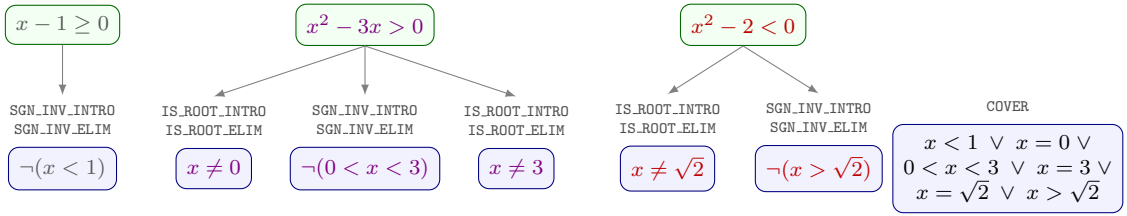
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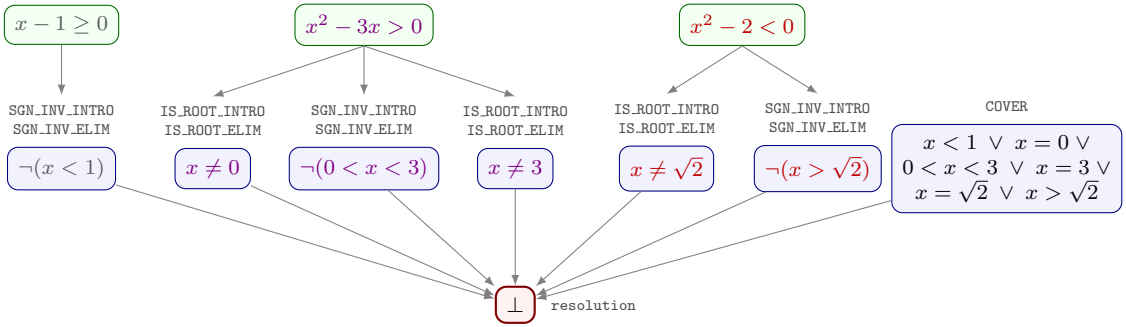
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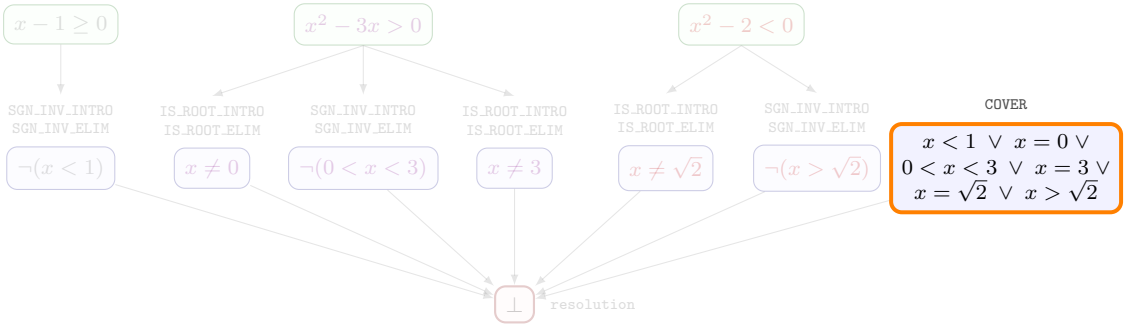


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- `COVER` receives the intervals as a parameter and has, as side condition, that they cover $\mathbb{R}$. The conclusion is that any $x \in \mathbb{R}$ must belong to at least one of the intervals.

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1: **function** $\textsc{sweep}(I_1, \ldots, I_k)$ ▷ The intervals are sorted by cvc5 by lower endpoint
2: $C \leftarrow \varnothing$ ▷ prefix of the line covered so far
3: **for** $j \leftarrow 1$ **to** k **do**
4: **if** $I_j = \{a\}$ **and** $C = (-\infty, a)$ **then**
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6: **else if** $I_j = (l, r)$ **and** $l \in C \cup \{-\infty\}$ **and** $r \notin C$ **then**
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1: **function** SWEEP($I_1, \ldots, I_k$) ▷ The intervals are sorted by cvc5 by lower endpoint
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```
theorem cover_of_sweep (L : List Interval) (h : sweep L = true) (x : Real) :
    orN (L.map (fun p => x ∈ p))
```

Real Algebraic Numbers

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- cvc5 (and our Lean module) represents RANs as a triple $\langle p, l, r \rangle$, where:
  - $p \in \mathbb{Q}[x]$ is a polynomial with rational coefficients, and
  - $l, r \in \mathbb{Q}$ are rationals such that p has a **unique** root in (l, r). The triple represents this root.

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- To compare RANs, cvc5 shrinks the intervals (l, r) until those of different numbers are disjoint, so `sweep` only needs to compare the rational bounds:

$$1 \ < \ \sqrt{2} = \langle x^2 - 2,\ \tfrac{5}{4},\ \tfrac{3}{2} \rangle \ < \ 3 \qquad \text{since} \qquad 1 < \tfrac{5}{4} \ \text{ and } \ \tfrac{3}{2} < 3$$

Well-Defined RANs

- We also need to cast our representation of RANs into real numbers in Lean. For that we need to **prove** that they are well defined, i.e., for a RAN $\langle p, l, r \rangle$, that there is indeed a single root of p in (l, r).

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- The same necessity will show up on the rule that excludes open intervals: we will need to show that p has **no** root in (l, r)

Sturm's Theorem

Sturm sequence

Given $p \in \mathbb{Q}[x]$, its **Sturm Sequence** $\mathrm{SturmSeq}(p)$ is

$$[p_0,\ p_1,\ p_2,\ \ldots,\ p_k],$$

where $p_0 := p$, $p_1 := p'$, $p_{i+1} := -\operatorname{rem}(p_{i-1}, p_i)$, and the sequence stops once $p_{k+1} = 0$.

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Let $\mathrm{Var}(seq, a)$ denote the number of sign changes of seq evaluated at a. If $a < b$ and $p(a), p(b) \neq 0$, then

$$|\{x \in (a,b) : p(x) = 0\}| \ = \ \mathrm{Var}(\mathrm{SturmSeq}(p), a) - \mathrm{Var}(\mathrm{SturmSeq}(p), b).$$

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Example

Is $\sqrt{2} = \langle x^2 - 2,\ 1,\ 2 \rangle$ well defined? $\quad \mathrm{SturmSeq}(x^2 - 2) = [\, x^2 - 2,\ 2x,\ 2 \,]$

	$x^2 - 2$	$2x$	2	sign changes
at 1	-1	2	2	1
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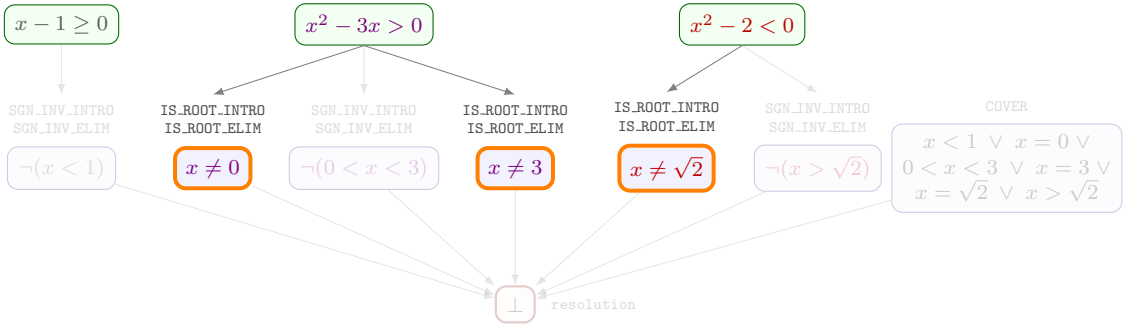
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- With rational a and b, this only needs rational arithmetic.
- cvc5 includes the Sturm sequences in the proof, so Lean's kernel only checks them instead of recomputing them.

The Shape of the Proof



Excluding Point Intervals

- Every point interval $\{\alpha\}$ will always be paired with an offending constraint $p(x) \bowtie 0$ such that $p(\alpha) = 0$ and $\bowtie$ is either $\neq$, $<$ or $>$.

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- To exclude such intervals we just need to confirm that $p(\alpha) = 0$, then we can use the offending constraint to conclude that $x \neq \alpha$.
  - If α is rational, we just evaluate $p(\alpha)$ and check that it is 0.
  - If $\alpha = \langle q, l, r \rangle$, then $q(\alpha) = 0$. Most of the time q will be chosen from the constraint itself, so it will be equal to p. If not, we rely on a variant of Sturm's theorem to compute the sign of $p(\alpha)$.

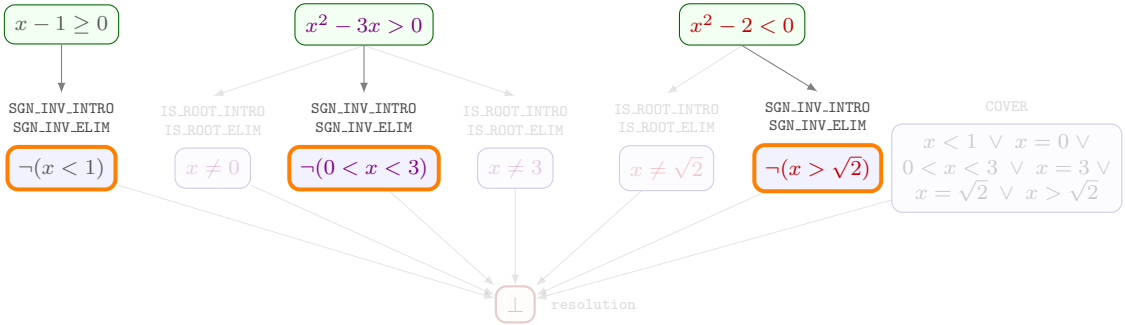
Excluding Point Intervals

- Every point interval $\{\alpha\}$ will always be paired with an offending constraint $p(x) \bowtie 0$ such that $p(\alpha) = 0$ and $\bowtie$ is either $\neq$, $<$ or $>$.
- To exclude such intervals we just need to confirm that $p(\alpha) = 0$, then we can use the offending constraint to conclude that $x \neq \alpha$.
  - If α is rational, we just evaluate $p(\alpha)$ and check that it is 0.
  - If $\alpha = \langle q, l, r \rangle$, then $q(\alpha) = 0$. Most of the time q will be chosen from the constraint itself, so it will be equal to p. If not, we rely on a variant of Sturm's theorem to compute the sign of $p(\alpha)$.

Example: $\{\sqrt{2}\}$ and $x^2 - 2 < 0$

$\sqrt{2} = \langle x^2 - 2,\ \frac{5}{4},\ \frac{3}{2} \rangle$. We already know that $\sqrt{2}$ is a root of $x^2 - 2$ since it is well-defined, so we can conclude $x \neq \sqrt{2}$ from $x^2 - 2 < 0$.

The Shape of the Proof



Excluding Open Intervals

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Example: $(0, 3)$ and $x^2 - 3x > 0$

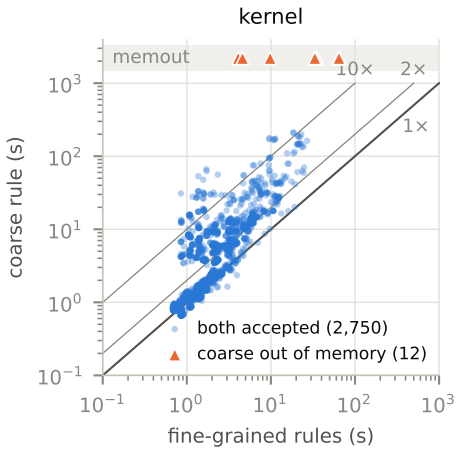
The polynomial $x^2 - 3x$ has no roots in $(0, 3)$. At $s = 1$, $p(s) = -2 \leq 0$.
Thus $x \notin (0, 3)$.

Fine-grained vs. Coarse-grained

- The coarse-grained checker is a reimplementation of the work of Li et al.

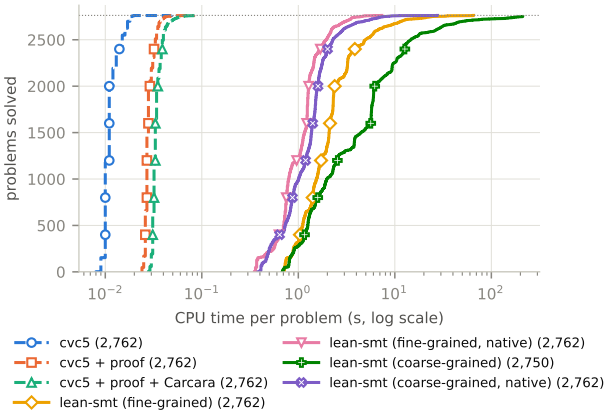
	Coarse-grained (old)	**Fine-grained (new)**
Certificate	all roots of all polynomials	intervals and their constraints
Regions	intervals between roots	intervals from coverings
False constraint on each region	found by the checker	given by cvc5
Sturm sequences	computed by the checker	given by cvc5
Signs at irrational points	Sturm–Tarski	Not needed / Sturm–Tarski

Fine-grained vs. Coarse-grained



▷ Checking in Lean's kernel is $3\times$ faster in total. In 12 the old checker runs out of memory.

All configurations



▷ Number of problems checked within a given time per problem, for each configuration

A complete Lean tactic

- None of `sos`, `nlinarith`, `grind` and `positivity` closes these:

```
-- OK, 2.7s
example (x : Real) (h1 : x*x*x - 3*x + 1 = 0) (h2 : 0.35 < x) (h3 : x < 1.5) :
    False := by smt [h1, h2, h3]

-- OK, 2.1s
example (x : Real) (h : x*x*x*x*x - x - 1 = 0) : 1.16 < x := by smt [h]

-- OK, 1.3s
example (x : Real) (h : x*x*x = 2) : 1.25 < x := by smt [h]

-- OK, 2.9s
example (x : Real) (h1 : 0 ≤ x) (h2 : x ≤ 2) :
    (x*x*x - 3*x) * (x*x*x - 3*x) ≤ 4 := by smt [h1, h2]
```

Conclusion and Future Work

- We presented a new framework for verifying the answers of the Cylindrical Algebraic Coverings algorithm for univariate problems.
- Our checker significantly outperforms the previous one, which was based on the state-of-the-art approach proposed by Li et al.
- Descartes' rule of signs could be used to speed up the verification of well-definedness of RANs.
- We are working on the multivariate case. We already formalized a significant part of the theory in Lean using Claude. We are now working on instrumenting the algorithm to produce proofs.